

Single-Panel Model Egitim Raporu

In-Silico Skorlar OLMADAN (v3_no_insil)

Tarih: 2026-03-24 14:56

Modeller: LightGBM, XGBoost, NN, DNN, SVM

Egitim: General / Hereditary_Cancer | Test-only: PAH / CFTR

1. Veri Ozeti

Toplam veri: 4287 satir

Feature sayisi (FE sonrasi): 952 (target ve Panel haric)

Panel Dagilimi:

General: 3156 satir (pos=2182, neg=974)

Hereditary_Cancer: 715 satir (pos=354, neg=361)

PAH: 324 satir (pos=323, neg=1) (test-only)

CFTR: 92 satir (pos=92, neg=0) (test-only)

2. Hold-Out Sonuclari

Panel	Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Sec
General	lightgbm	0.9465	0.9674	0.9825	0.9219	0.9725	0.8192	24
General	xgboost	0.9487	0.9686	0.9851	0.9259	0.9725	0.8269	41
General	nn	0.8814	0.8813	0.9338	0.8621	0.9016	0.5972	131
General	dnn	0.8886	0.8819	0.9340	0.8655	0.9130	0.6187	47
General	svm	0.8955	0.8960	0.9436	0.8623	0.9314	0.6352	149
Hereditary_Cancer	lightgbm	0.9362	0.9810	0.9822	0.9429	0.9296	0.8742	8
Hereditary_Cancer	xgboost	0.9343	0.9806	0.9809	0.9697	0.9014	0.8762	29
Hereditary_Cancer	nn	0.8414	0.8865	0.8931	0.8243	0.8592	0.6790	23
Hereditary_Cancer	dnn	0.8593	0.9018	0.9064	0.9062	0.8169	0.7376	7
Hereditary_Cancer	svm	0.8800	0.9235	0.9257	0.8354	0.9296	0.7532	7

En İyi Model (Panel Bazında):

General: xgboost (F1=0.9487)

Hereditary_Cancer: lightgbm (F1=0.9362)

3. Test-Only Panel Sonuclari (General modeli ile)

Panel	Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Bal.Acc	N
CFTR	lightgbm	0.9721	0.0000	0.0000	1.0000	0.9457	0.0000	0.9457	92
PAH	lightgbm	0.9427	0.9505	0.9998	1.0000	0.8916	0.1574	0.9458	324
CFTR	xgboost	0.9721	0.0000	0.0000	1.0000	0.9457	0.0000	0.9457	92
PAH	xgboost	0.9340	0.9845	1.0000	1.0000	0.8762	0.1462	0.9381	324
CFTR	nn	0.9302	0.0000	0.0000	1.0000	0.8696	0.0000	0.8696	92
PAH	nn	0.8667	0.9783	0.9999	1.0000	0.7647	0.0997	0.8824	324
CFTR	dnn	0.9176	0.0000	0.0000	1.0000	0.8478	0.0000	0.8478	92
PAH	dnn	0.8726	0.9814	0.9999	1.0000	0.7740	0.1023	0.8870	324
CFTR	svm	0.9663	0.0000	0.0000	1.0000	0.9348	0.0000	0.9348	92
PAH	svm	0.9215	0.9505	0.9998	1.0000	0.8545	0.1334	0.9272	324

En iyi Model (Test-Only Panel Bazinda):

CFTR: lightgbm (F1=0.9721, Recall=0.9457)

PAH: lightgbm (F1=0.9427, Recall=0.8916)

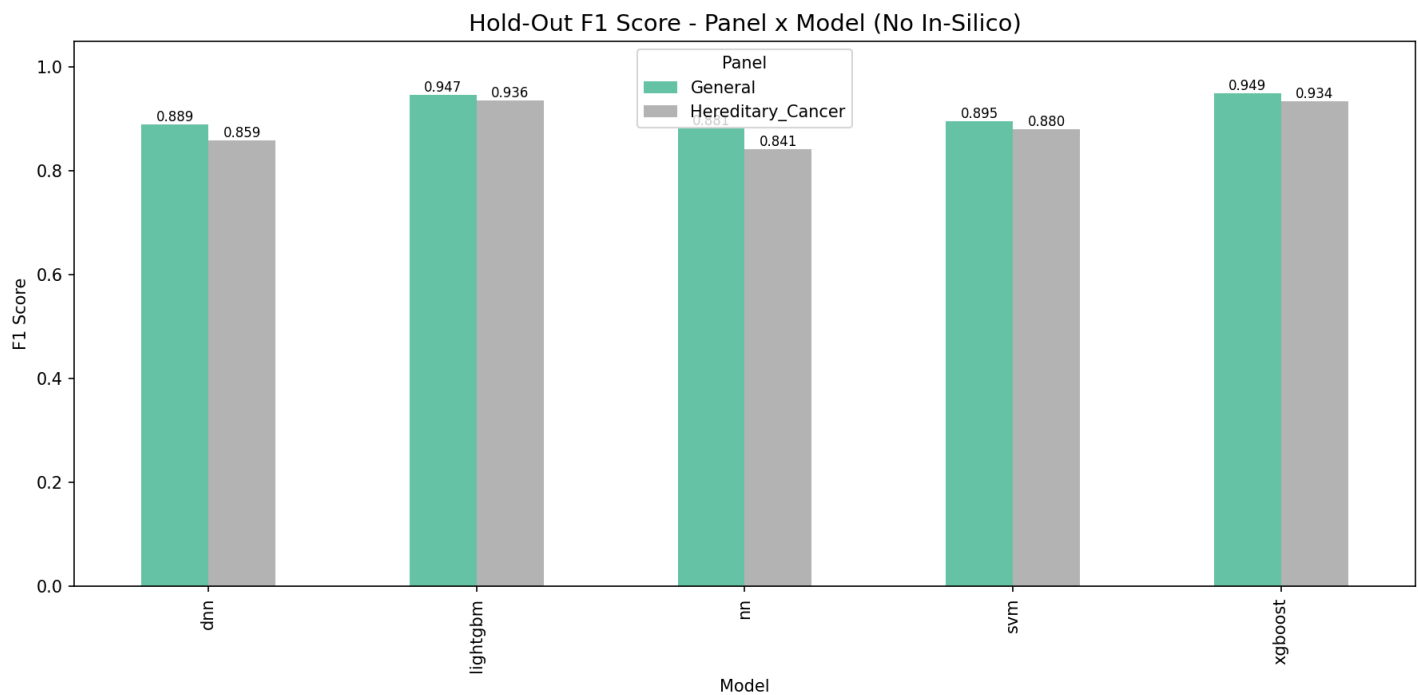
4. Cross-Model Error Analizi

General: 632 test satiri | Tum yanlis: 18 | Cogunluk yanlis: 82 | Tum dogru: 480

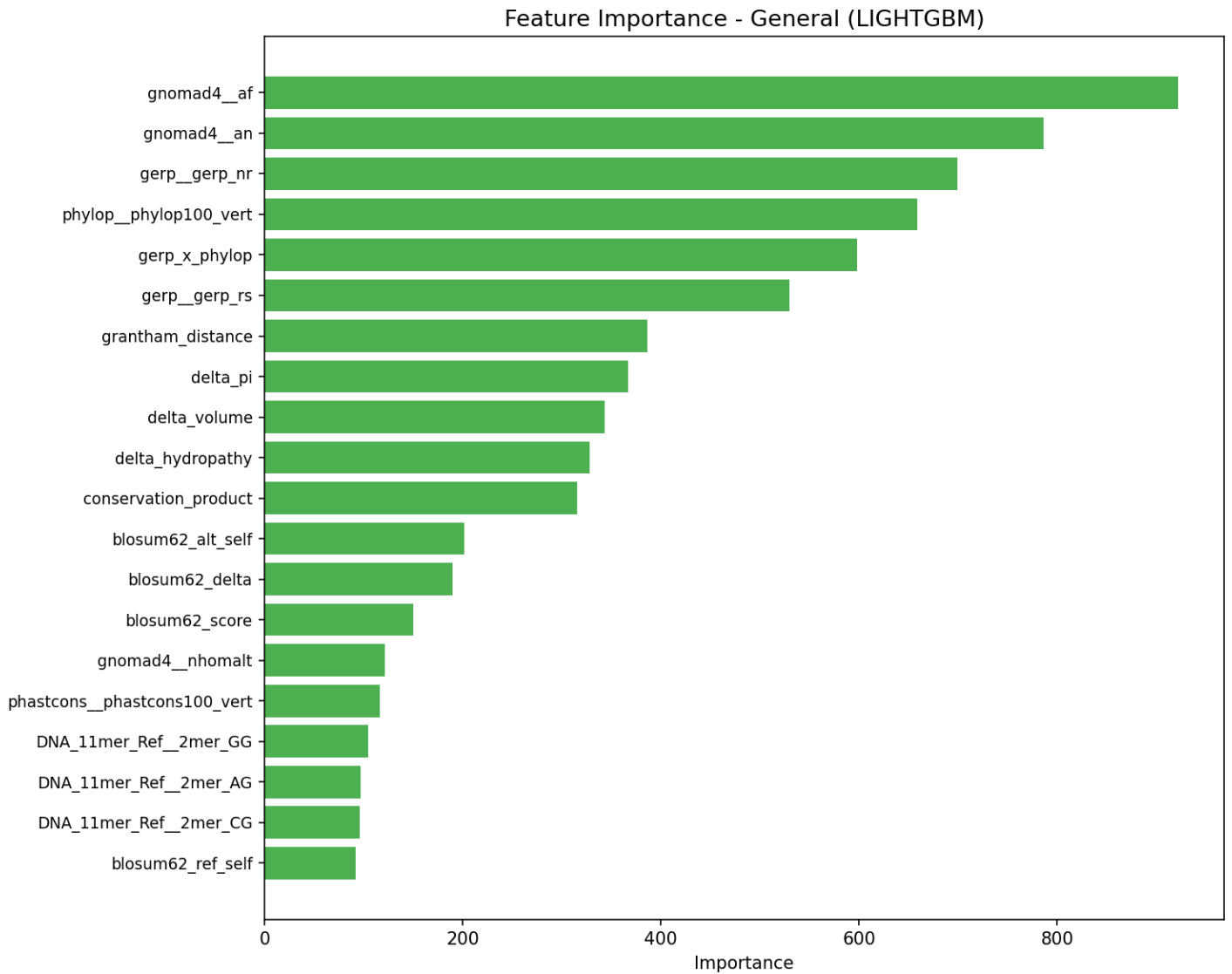
Hereditary_Cancer: 143 test satiri | Tum yanlis: 0 | Cogunluk yanlis: 15 | Tum dogru: 106

Not: Tum modellerin yanlis buldugu satirlar ground truth acisindan supheli olabilir.

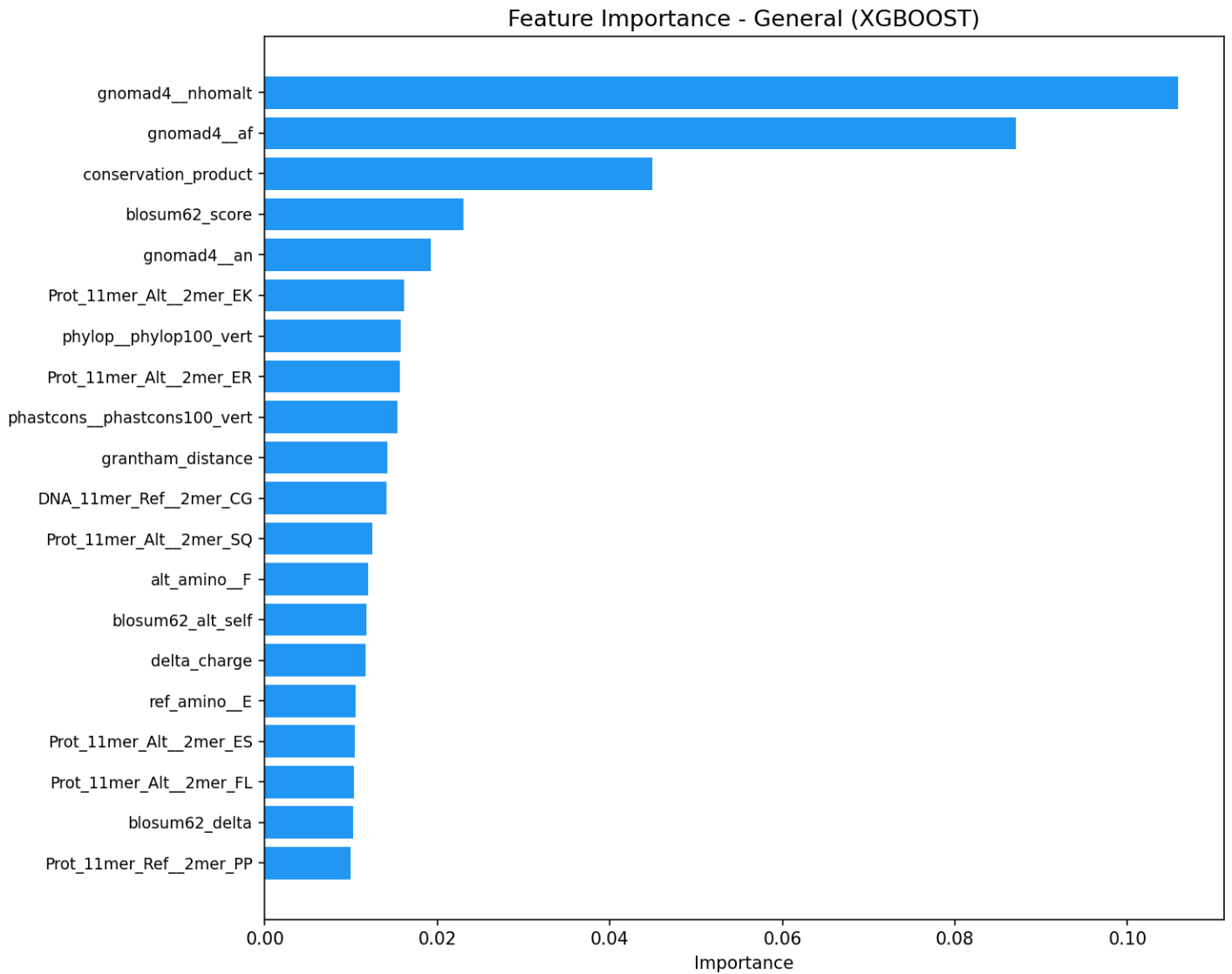
F1 Comparison



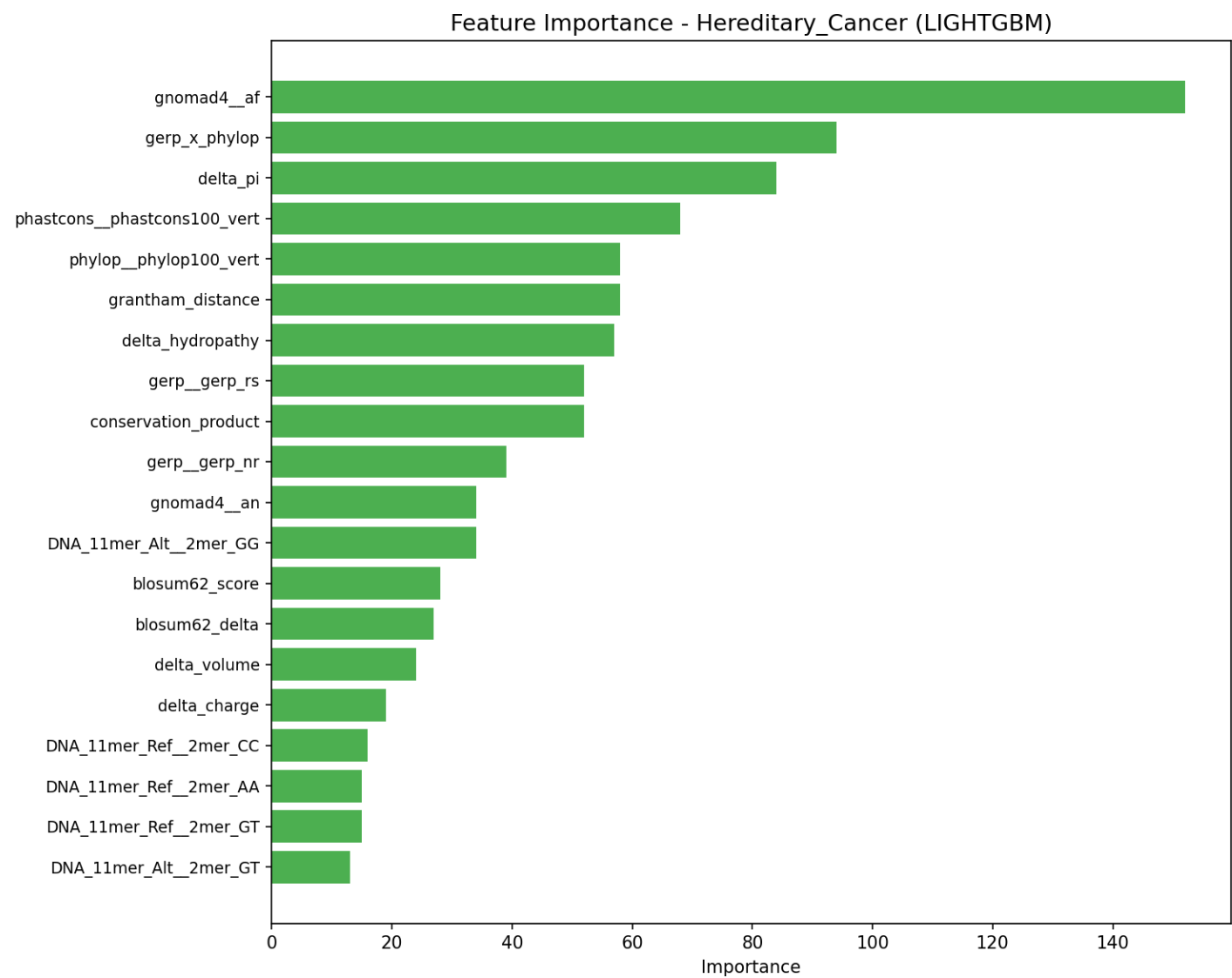
General Lightgbm Feature Importance



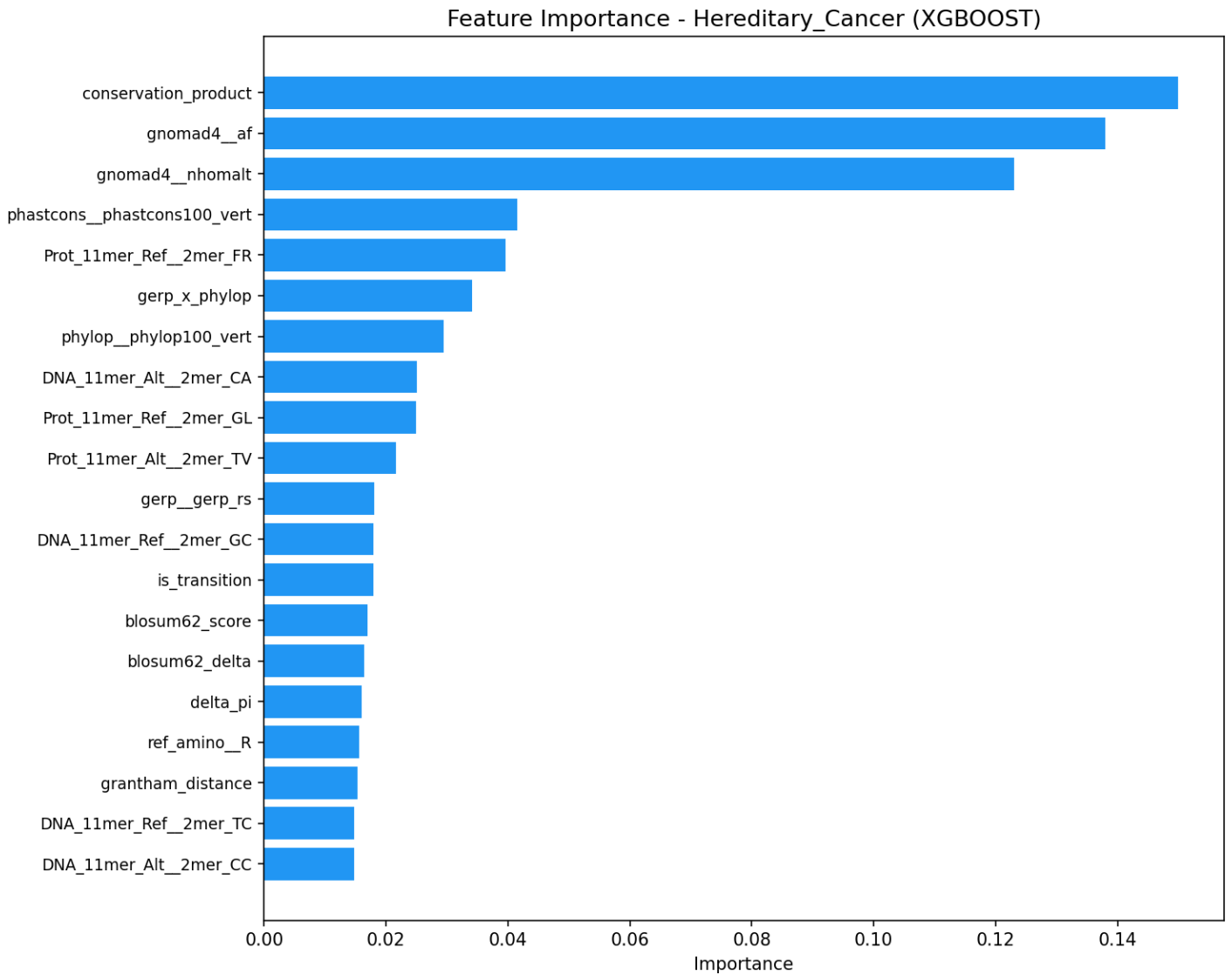
General Xgboost Feature Importance



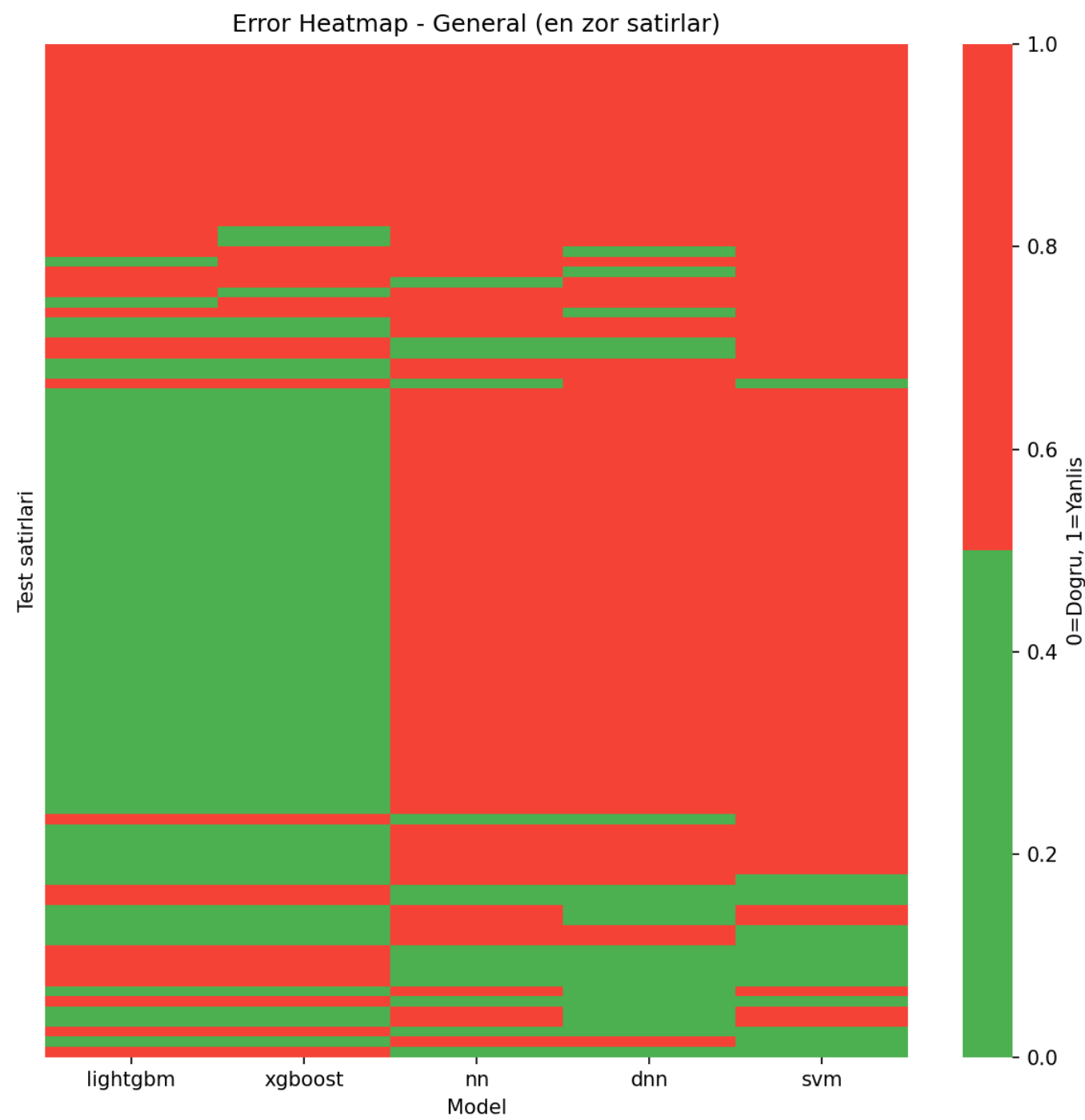
Hereditary Cancer Lightgbm Feature Importance



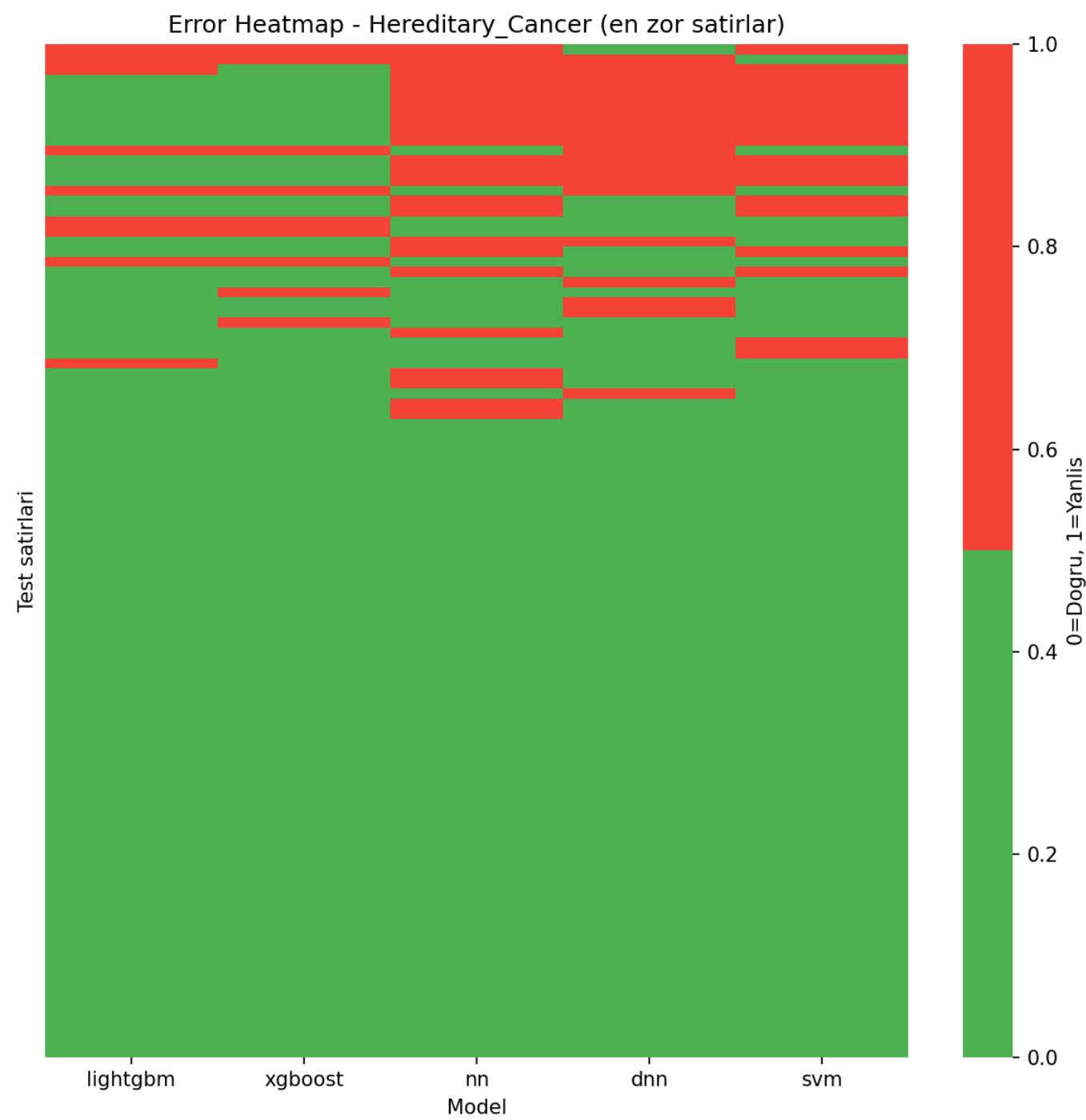
Hereditary Cancer Xgboost Feature Importance



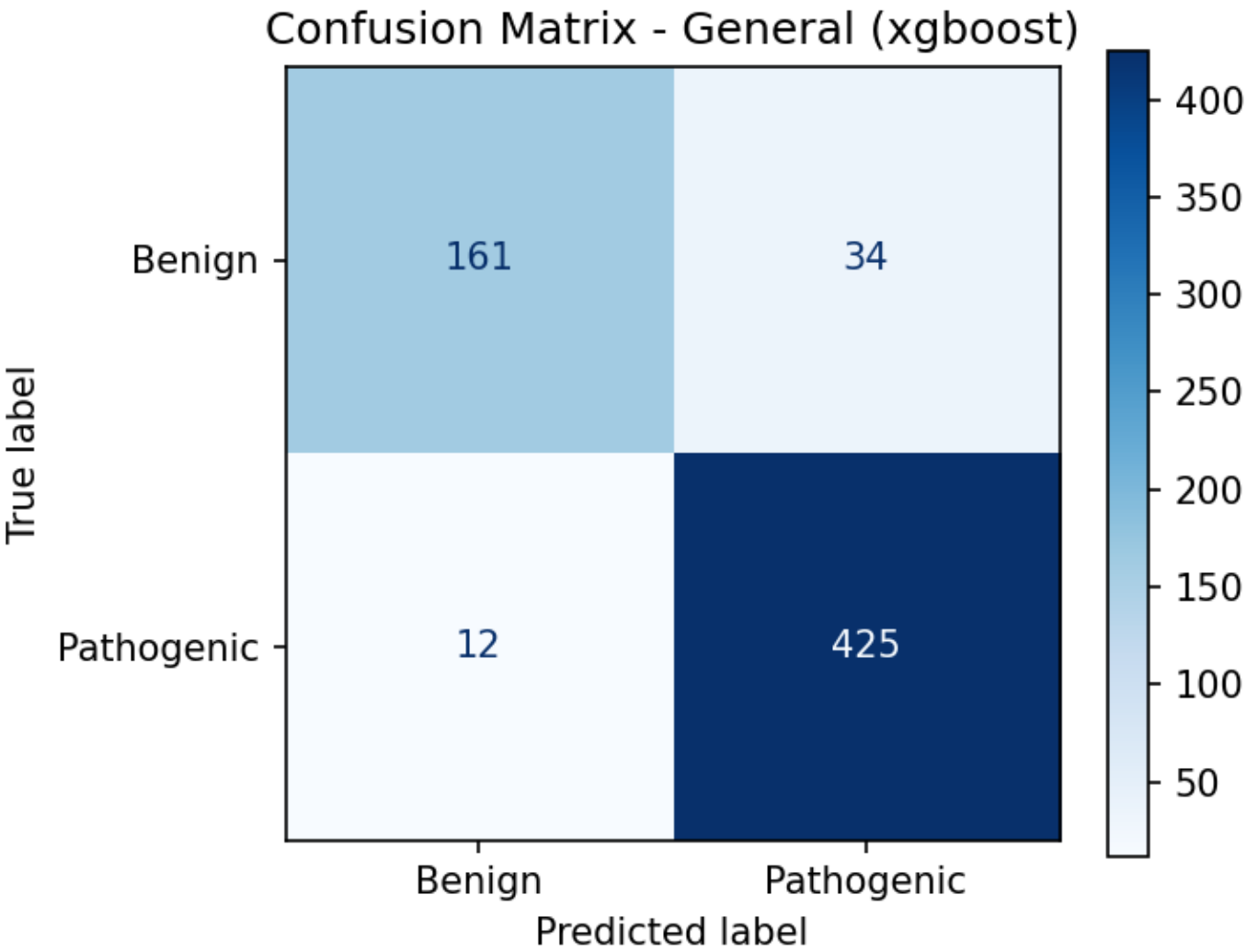
General Error Heatmap



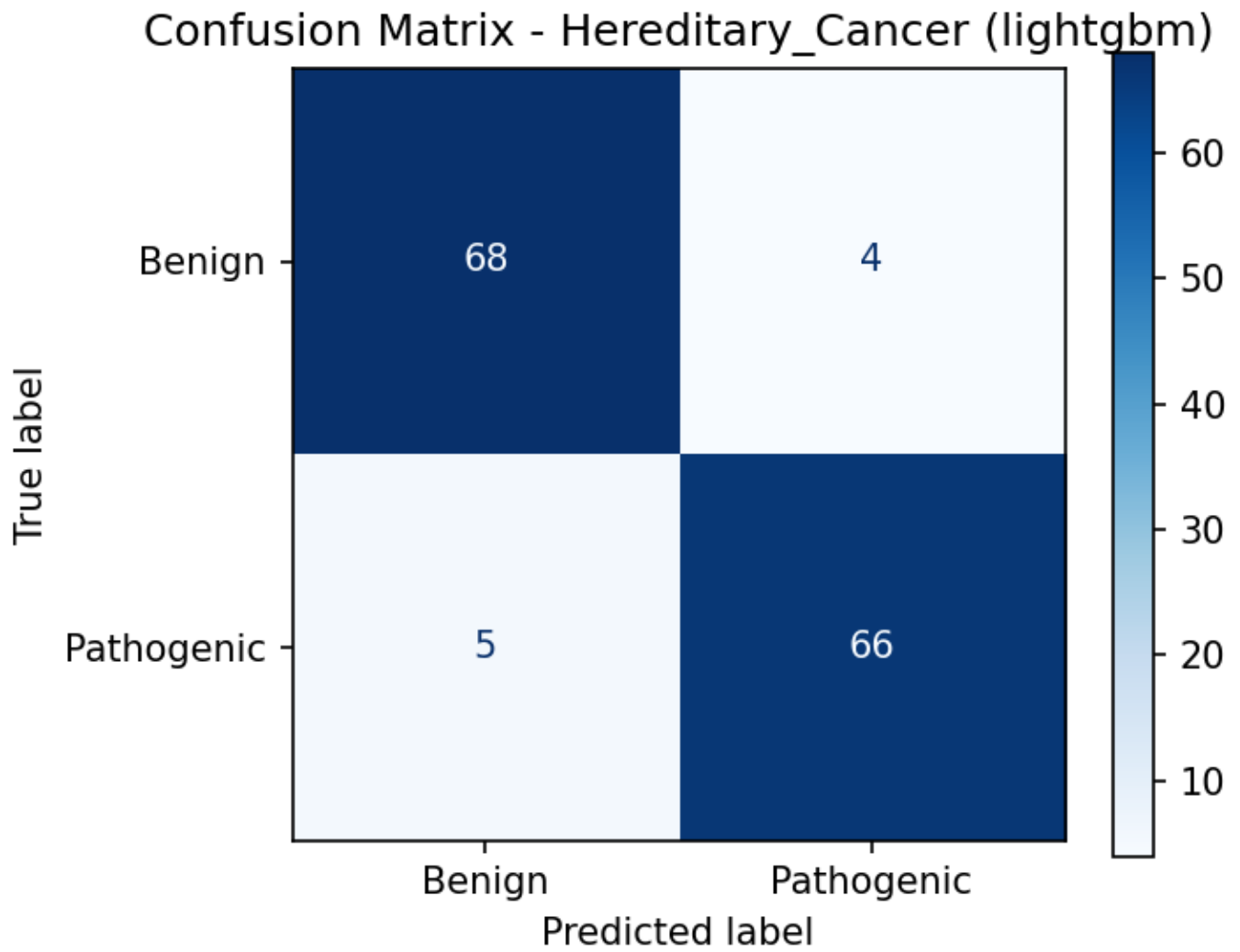
Hereditary Cancer Error Heatmap



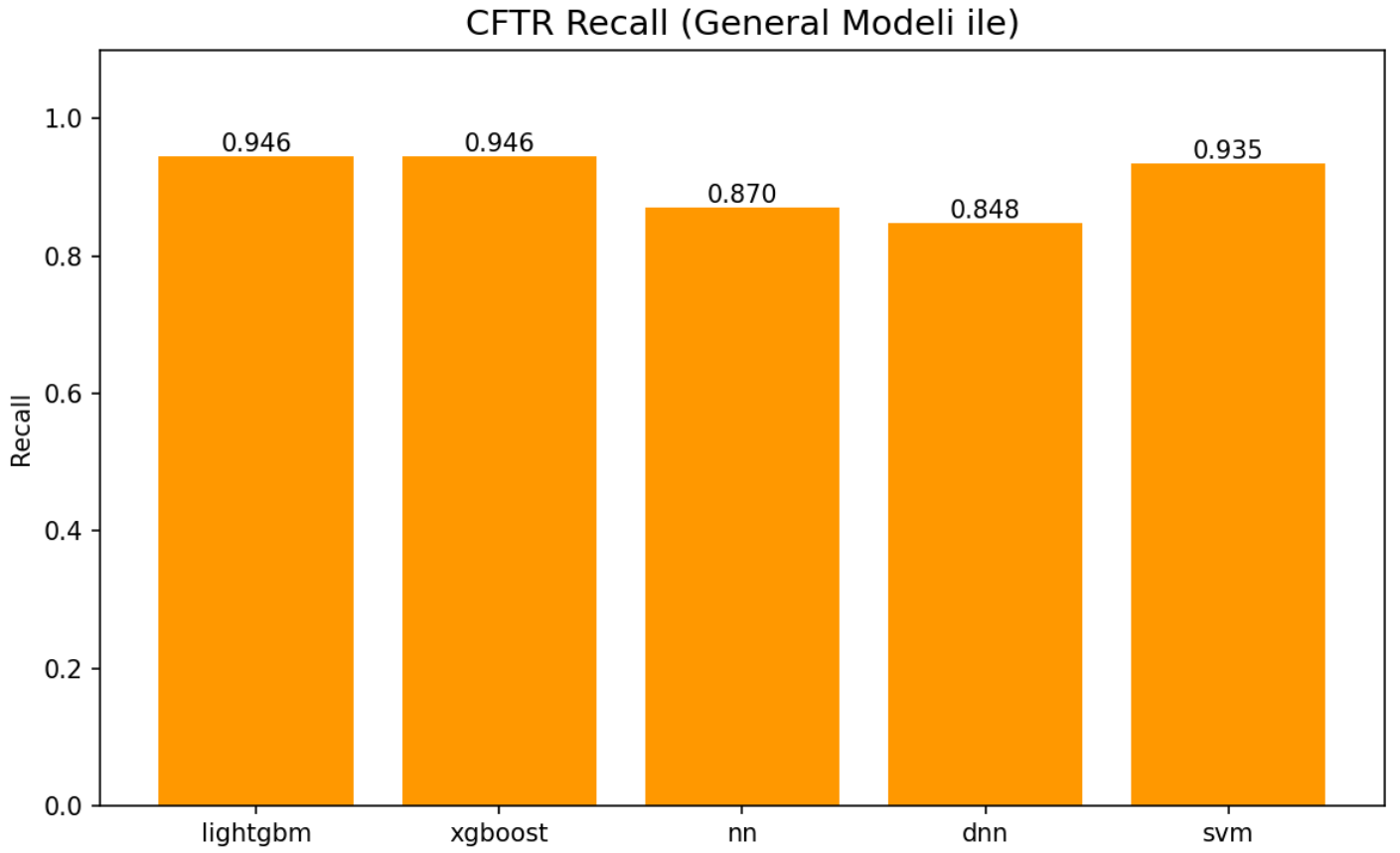
General Confusion Matrix



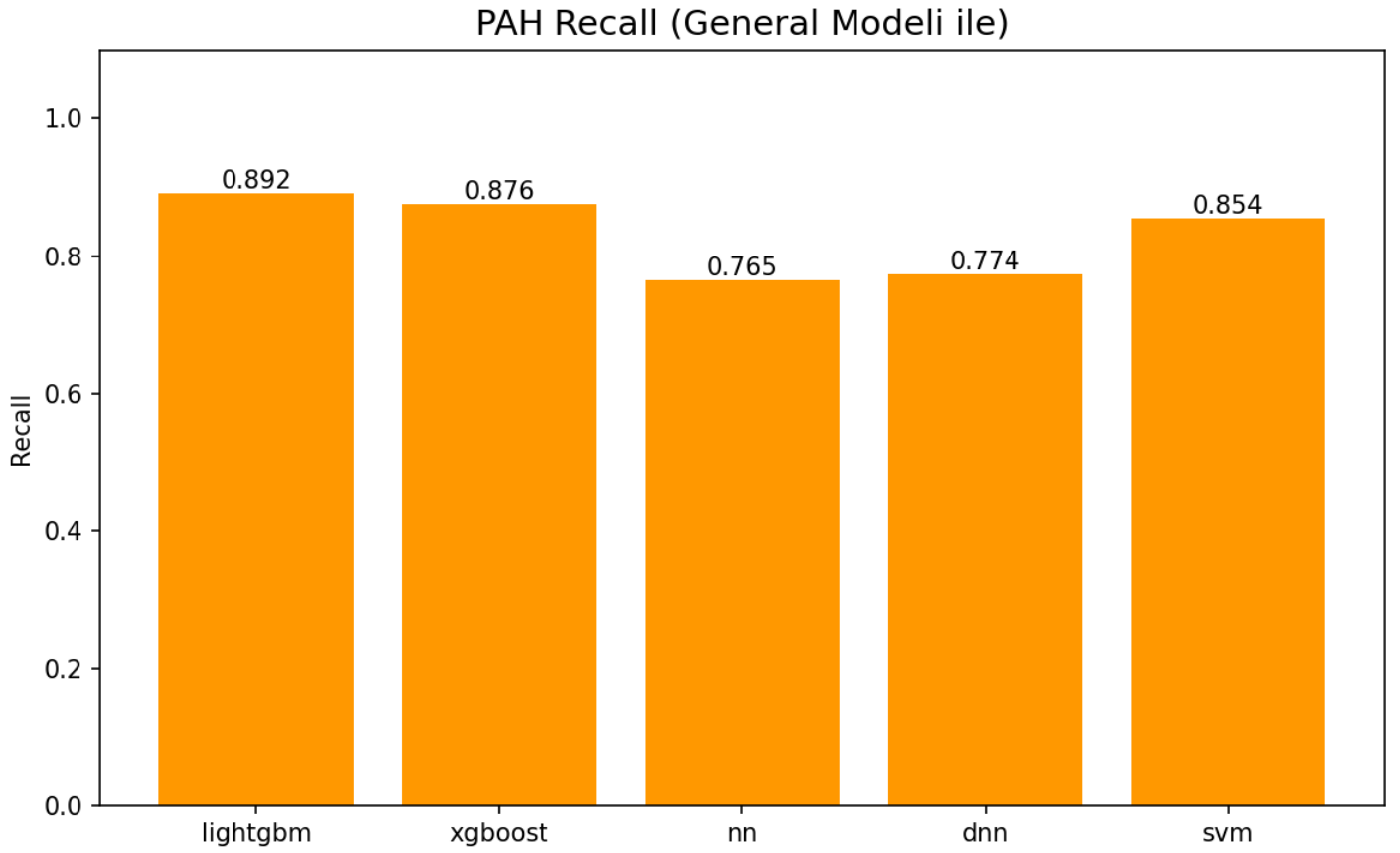
Hereditary Cancer Confusion Matrix



Cftr Recall



Pah Recall



Difficulty Distribution

